Choices so far

System One Control of LLM Agent Harnesses with Verifiable Rewards. Companion to `results_plan.pdf`. Last updated 3 November 2026.

Halil's choices stand until he changes them. Claude's are proposals until he accepts them, and accepting one moves it to the first table. Later sessions add a row here whenever a choice is made, and do not present anything outside the first table as Halil's.

1 Chosen by Halil

#	Topic	The choice	Date
H1	Target	Make Gemma-4-E4B match about Qwen3.8-27B in success, with a lightweight trained Sys1	3 Nov
H2	Sys1's job	At each step pick from a menu of frequent actions, or abort, flag danger, or fall back to the LLM. Control returns when the LLM finishes	1 Nov
H3	Menu	Built statistically from the most common commands. Its length is a hyperparameter tuned by performance	1 Nov
H4	Hand-over rules	After repeated fallbacks the LLM keeps control for a number of rounds, and a run of simple LLM calls returns control. The numbers are hyperparameters	1 Nov
H5	What Sys1 sees	The LLM's last tool trace, and the harness control state	1 Nov
H6	Reward	Match a strong model's choices as classification. When its choice is off the menu, fallback is the correct pick	1 Nov
H7	Teacher forcing	Used to amplify the learning signal in early stages. The rule for whose action runs can be a research path	1 Nov
H8	Base model	Any open-weight, already trained System One model, trained further, for example Cloudflare's. Not bound to Jev	1 Nov
H9	What counts as success	Faster, cheaper, more accurate or more secure than the LLM alone. Any one is enough. Negative results are acceptable	1 Nov
H10	Security	Simpler subtasks, possibly a different task for the guardrail part	1 Nov
H11	Tasks	τ-bench was only an example. Stay in line with the comparison papers	3 Nov
H12	Compute	Colab L4, about 600 hours a month, or G4. One person	1 Nov
H13	Scope cuts	Drop training parts, simplify tasks, and as a last resort a System One tool selector the agent calls	1 Nov

2 Proposed by Claude, waiting for Halil

#	Proposal	Why	In the plan
C1	ALFWorld as the primary task	Bonn reports Gemma and Qwen 27B on these exact 134 games, and ReDAct also reports on ALFWorld	Tasks table
C2	MiniGrid DoorKey-8x8 as second task	ReDAct's setup, and cheap for RL	Tasks table
C3	AgentDojo for the guardrail, CyberGym optional	H10. AgentDojo has labelled injections and programmatic checks, and CyberGym needs a Docker host	Q6
C4	ScienceWorld as extension	SwiftSage reports on it	Tasks table
C5	Kev-4B as primary Sys1, decider-2b and Clef-flash as alternatives	Kev has a documented training path and no per-option parameters. H8 names Cloudflare	Models table
C6	Five training arms A0 to A4	They walk along H7's research path	Arms table
C7	Rewards R1 to R4	R1 is H6. The others add costs or success	Below the arms
C8	Keep teacher labels only from games the checker marks solved	Makes "the teacher is right" a checked statement	Harness table
C9	At fallback, the LLM reasons and then gives a command	Fallback is for steps that need reasoning	Harness table
C10	The target line is the stronger thinking-off Qwen 27B row, and thinking on is a ceiling	Bonn's 54.5 has thinking off, Qwen 27B thinks by default, and D6 lets you change this	Q1, Q4
C11	Reproduce Bonn's numbers before anything else	Otherwise our rows cannot sit beside theirs	G
C12	Tune on dev, test once, 3 seeds, paired bootstrap	Standard practice, and Bonn's games make pairing possible	Below the arms
C13	Tier 1 and Tier 2 ablations, and the cut order	H13	Q7, Budget
C14	Draft fail conditions and what each no would mean	So "no" is defined before data exists. Thresholds are guesses	Draft section
C15	Replay scores Gemma's stored one-pass pick during LLM rounds, as a per-step proxy	Otherwise fallback looks like expert play in training and like Gemma at test (agent review)	Harness table
C16	Row H11: Gemma fine-tuned to pick the expert's command, no Sys1	Fine-tuning is the usual practice on ALFWorld (ETO, AgentTuning, AgentGym), and Sys1 should have to beat it	Q4
C17	Kev-0.8B as an option, and every result states its precision	Gemma bf16 plus Kev-4B does not fit an L4	Fit table
C18	`important_instructions` as the main AgentDojo attack	The paper's main attack, where `tool_knowledge` is only the README's example	Q6
C19	The menu scores each admissible command by teacher use with that object and place class	Ranking by kind alone could fill the menu with `go to` (agent review)	Harness table
C20	RL arms start from A1's weights	decider, RelayLLM and ETO start RL from supervised weights	Arms table
C21	The no-hand-over row moves to Tier 1	The papers read hand over one step or one move, so H4 is what is distinctive	Q7

3 Hyperparameters

Name	What it sets	Start	Values tried	Start value from
Menu length k	How many commands Sys1 sees	10	5, 10, 20, all	Claude. Tuning is H3
Fallbacks before hand-over	How many in a row give the LLM control	2	1, 2	Halil
LLM rounds after hand-over	How long the LLM keeps control	10	5, 10	Halil
Menu run that returns control	LLM commands in a row, all on the menu	3	3	Claude
Trace window	Trace entries Sys1 sees	5	1, 5, 20, full	Claude
Wrong-pick cost w	Reward lost on a wrong pick	1	0, 1	Claude
Needless-fallback cost c	Reward lost on fallback when the teacher's command was on the menu	0.1	0, 0.1	Claude
Success bonus β	Reward at a successful end	0	0, 1, 5	Claude
Teacher schedule (A3)	How fast the teacher's chance of acting falls	1 to 0	Shape open	Claude
Teacher games	Games stored as training labels	3,553 with the expert, 500 with a paid LLM	Open	Claude
Handovers per game (Q5)	Steps per game that may go to Qwen 27B	5	1, 2, 5, 10	ReDAct's target
PPO and optimizer settings	Learning rate, batch, clip and others	Open	Set at the first run. decider used 1e-6 and clip 0.2	Not chosen
RL training budget	Steps per RL arm	Open	Field range: decider 384 optimizer steps, TWOSOME up to 500k, GLAM 1.5M, on other tasks	Not chosen

4 Open decisions

Table 1: Each decision changes what runs. Claude's suggestion is stated, and the plan works under any option.

#	Decision	Options and what each costs	Claude's suggestion
D1	Who is the teacher?	(a) Birch through your cloud credits, as you planned, with collection before 5 November. Its provider's commercial and consumer terms restrict using the service to train competing models and restrict automated access outside an API key. Reading them is yours, and Claude gives no legal opinion. (b) ALFWorld's built-in expert and our MiniGrid solver, free at any state. (c) An open strong LLM such as Qwen 27B	(b) for all arms, because A3 and A4 need a label at every state Sys1 reaches. This departs from your strong LLM as the default teacher. Then (a) or (c) as the teacher rows in the Q7 table, which keeps your strong-LLM question in the results
D2	Where do abort and flag danger get their labels?	An agreement reward never labels them, because the teacher never picks them. (a) Flag danger trained on AgentDojo's labelled injections, abort as a hand rule on repeated commands. (b) Abort and flag danger with a cost in the reward. (c) Your own rule	(a)
D3	Is ALFWorld acceptable as the primary task?	Your target has published numbers on it and it is free to check, but its commands are not tool calls. The alternative is a tool-calling primary such as τ retail, which brings a paid user simulator on every turn and has no published Gemma or Qwen 27B numbers that we read	ALFWorld, with AgentDojo as the tool-calling evidence
D4	Does security rest on AgentDojo alone?	CyberGym measures exploitation capability and needs a Docker host and a large download. Its 10-task subset stays as counts	AgentDojo as the core, CyberGym only if a Docker host is at hand
D5	What runs together on our GPU?	See the fit table in the plan. (a) L4 with Gemma bf16 and Kev-0.8B, and Qwen 27B in 4-bit, stated as a different setup from Bonn's. (b) L4 with Gemma 8-bit and Kev-4B, which moves Gemma away from Bonn's setup. (c) G4, whose memory we read at setup. (d) No live Qwen 27B: Bonn's number is the target line, and H8, H10, H12 and Q5's Qwen 27B rows drop out	Measure (a) in the first run, then choose
D6	Which Qwen 27B is the bar?	Bonn's 54.5 has thinking off, and Qwen 27B thinks by default. (a) The stronger thinking-off row. (b) Bonn's published 54.5 only. (c) Thinking on, which raises the bar and the running time	(a), with thinking on reported as a ceiling

5 Draft fail conditions, by Claude

Table 2: When an answer counts as a no, and what that no would mean. Claude's draft, so that a no is defined before any data exists. Halil did not write it and should accept or rewrite it. Thresholds are guesses.

#	Counts as a no when	What that no would mean
G	Our success falls outside Bonn's interval, or under 95% of first-step picks match Bonn's public responses, where the prompt is the same	Our setup differs from Bonn's. Fix it before any other run
Q1	The menu of length 20 holds under 50% of teacher commands	Ranking by frequency gives no useful menu, so Sys1 chooses from all admissible commands
Q2	Trained minus untrained, the interval on step match or ECE includes 0	The labels do not teach Sys1 at this budget, so inspect the teacher
Q3	RL minus A1, the interval on dev success includes 0, and the exposure gap does not shrink when Sys1 acts	Plain copying is enough here and control state adds no signal. Still a result about RL
Q4	Target missed: H7 minus the bar (D6), the interval's low end is below -10 points, which is the finest "about" 134 games can support. Any gain missed: H7 minus Gemma alone, the interval includes 0. Gemma adds nothing: H7 minus H4 includes 0. Sys1 adds nothing over fine-tuning: H7 minus H11 includes 0	Gemma + Sys1 does not reach Qwen 27B, does not help at all, does not need Gemma, or is matched by plain fine-tuning. Each is a different result. Q5 and Q6 still stand on their own
Q5	At 5 handovers per game, Sys1 minus Gemma's own uncertainty is at or below 0	A threshold with no training chooses the steps as well as trained Sys1
Q6	Attack success, Sys1 minus no defence, the interval includes 0, or task success falls further than with `tool_filter`	Reading from outside does not stop these attacks at an acceptable cost
Q7	Removing a part does not move success or cost in the expected direction. One seed cannot settle it	That part is a candidate to leave the story, pending a run with 3 seeds